

西南科技大学

2024-2025-1 学期《网络安全技术》

实验 1 实验报告

实验名称：网络资源信息收集

实验类型：验证性实验

指导教师：李凌

专业班级：

姓名：

学号：

小组组号：第 组

实验地点：个人计算机 .

实验日期：2024 年 09 月 18 日

实验成绩：_____

一、实验目的

- 掌握利用公开渠道收集目标系统信息的原理和手段；
- 了解互联网中可能给攻击者带来便利的公开信息；
- 通过使用 ARP 和 ICMP 协议对主机进行扫描探测，加深对原理的认识；
- 具备利用 Nmap 扫描器进行主机扫描探测和利用 Wireshark 工具进行数据包分析的能力；
- 加深对操作系统类型探测和端口扫描探测原理的认识；
- 掌握利用 Nmap 进行操作系统类型的探测和端口扫描方法。

二、实验设计

信息收集网站：不能使用西科大和百度。

搜索引擎、whois

实验环境：

攻击机 kali, ip 为 192.168.121.129, 自带 wireshark、nmap。

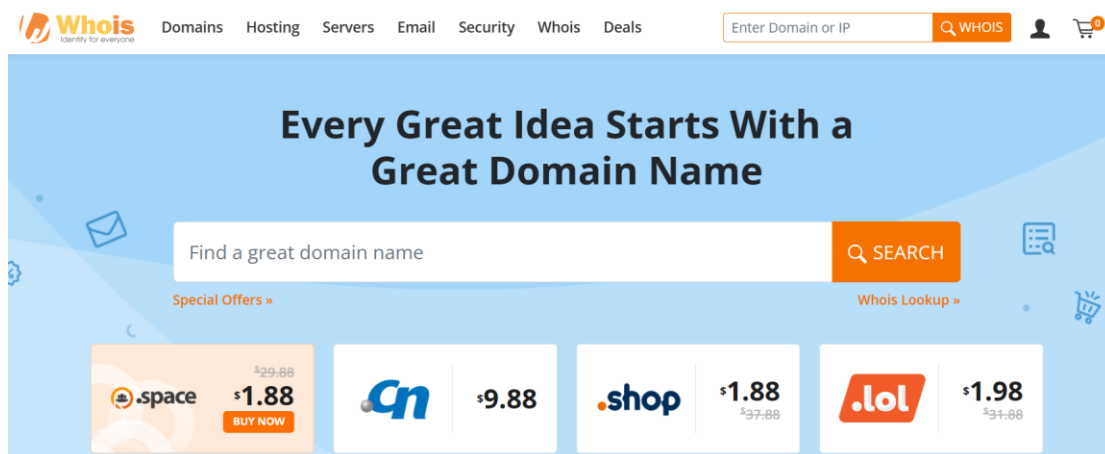
被攻击机 winxp(宿主机), ip: 192.168.121.130

win10（开放防火墙），ip: 192.168.121.128

三、实验过程（包含实验结果）

（1）使用 Baidu、Google 等等搜索引擎访问目标网站，并从中收集可能会对网站带来危害的公开信息。

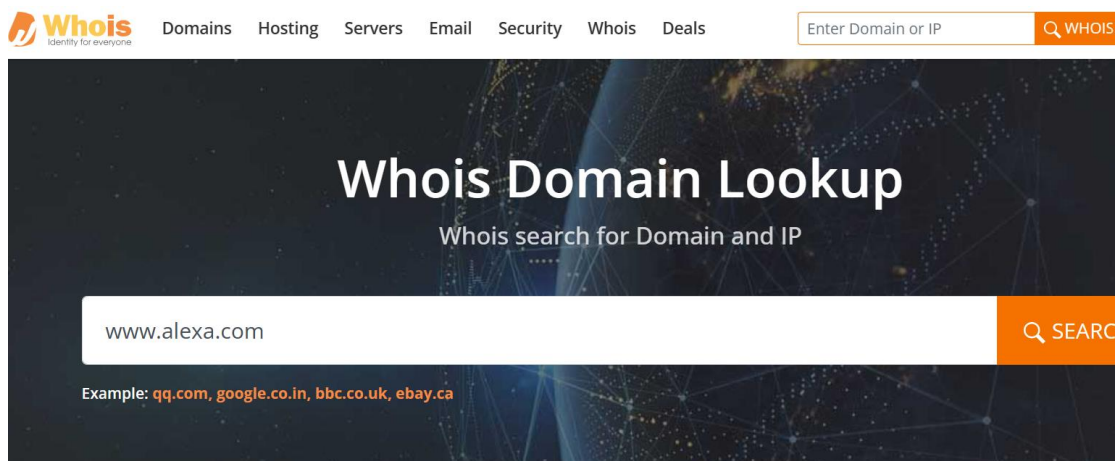
在导航栏上选择 Whois Lookup 标签



搜索 www.bing.com



搜索 www.alexa.com



有可能会对网站带来危害的公开信息：

1. 域名注册信息

注册人信息：注册人的姓名、地址、电子邮件和电话号码等信息，如果这些信息被恶意用户获取，可能会导致针对注册人的社交工程攻击或骚扰。

注册日期和到期日期：了解域名的注册和到期时间可以帮助攻击者判断域名是否容易被抢注，尤其是在到期时。

注册商信息：注册商的名称和联系方式，如果攻击者知道域名的注册商，可能会尝试通过伪造身份进行域名转移。

2. 域名解析服务器信息

DNS 服务器 IP 地址：公开的 DNS 服务器 IP 地址可能会被攻击者用来进行 DDoS 攻击或其他网络攻击。

DNS 记录类型：通过查询 DNS 记录（如 A 记录、MX 记录、NS 记录等），攻击者可以获取有关网站架构和邮件服务器的信息，这可能会被用于针对性的攻击。

子域名信息：通过 DNS 查询，攻击者可以发现与主域名相关的子域名，这些子域名可能存在安全漏洞，成为攻击的目标。

3. 其他潜在风险

过期域名的再注册：如果域名即将到期且未及时续费，攻击者可能会尝试抢注该域名，进而进行钓鱼攻击或其他恶意活动。

信息泄露：通过 Whois 查询，攻击者可以获取到的注册信息可能会被用于构建针对该网站的攻击策略，例如通过社交工程获取更多敏感信息。

（2）使用 Whois 服务查询网站的域名注册信息，以及对目标网站的域名解析服务器信息的收集

使用 Whois 服务查询网站的域名注册信息，以及对目标网站的域名解析服务器信息的收集：

www.bing.com

bing.com

Updated 3 days ago 



Domain Information

Domain:	bing.com
Registrar:	MarkMonitor Inc.
Registered On:	1996-01-29
Expires On:	2025-01-30
Updated On:	2023-12-29
Status:	clientDeleteProhibited clientTransferProhibited clientUpdateProhibited serverDeleteProhibited serverTransferProhibited serverUpdateProhibited
Name Servers:	dns1.p09.nsone.net dns2.p09.nsone.net dns3.p09.nsone.net dns4.p09.nsone.net ns1-204.azure-dns.com ns2-204.azure-dns.net ns3-204.azure-dns.org ns4-204.azure-dns.info



Registrant Contact

Name:	Domain Administrator
-------	----------------------



Domains

Hosting

Servers

Email

Security

Whois

Deals

Enter Domain or IP

State:	WA
Postal Code:	98052
Country:	US
Phone:	+1.4258828080
Fax:	+1.4259367329
Email:	domains @microsoft.com



Administrative Contact

Name:	Domain Administrator
Organization:	Microsoft Corporation
Street:	One Microsoft Way,
City:	Redmond
State:	WA
Postal Code:	98052
Country:	US
Phone:	+1.4258828080
Fax:	+1.4259367329
Email:	domains @microsoft.com

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 Domains Hosting Servers Email Security Whois Deals

Enter Domain or IP

Technical Contact

Name:	MSN Hostmaster
Organization:	Microsoft Corporation
Street:	One Microsoft Way,
City:	Redmond
State:	WA
Postal Code:	98052
Country:	US
Phone:	+1.4258828080
Fax:	+1.4259367329
Email:	msnhst @microsoft.com

Raw Whois Data

```
Domain Name: bing.com
Registry Domain ID: 49639_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2023-12-29T11:19:45+0000
Creation Date: 1996-01-29T05:00:00+0000
Registrar Registration Expiration Date: 2025-01-30T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
```

www.alexa.com

 Domains Hosting Servers Email Security Whois Deals

Enter Domain or IP

Q WHOIS

Whois Domain Lookup

Whois search for Domain and IP

www.alexa.com

Q SEARCH

Example: qq.com, google.co.in, bbc.co.uk, ebay.ca



Domains

Hosting

Servers

Email

Security

Whois

Deals

Enter Domain

alexa.com

Updated 1 day ago ↻



Domain Information

Domain:	alexa.com
Registrar:	MarkMonitor Inc.
Registered On:	1996-07-17
Expires On:	2025-07-16
Updated On:	2024-04-30
Status:	clientDeleteProhibited clientTransferProhibited clientUpdateProhibited serverDeleteProhibited serverTransferProhibited serverUpdateProhibited
Name Servers:	ns-115.awsdns-14.com ns-1324.awsdns-37.org ns-1556.awsdns-02.co.uk ns-672.awsdns-20.net



Registrant Contact

Organization:	Amazon Technologies, Inc.
State:	NV
Country:	US
Email:	Select Request Email Form at https://domains.markmonitor.com/whois/alexa.com

 Domains Hosting Servers Email Security Whois Deals

Email: [Select Request Email Form at https://domains.markmonitor.com/whois/alexa.com](https://domains.markmonitor.com/whois/alexa.com)

 **Administrative Contact**

Organization: Amazon Technologies, Inc.

State: NV

Country: US

Email: [Select Request Email Form at https://domains.markmonitor.com/whois/alexa.com](https://domains.markmonitor.com/whois/alexa.com)

 **Technical Contact**

Organization: Amazon Technologies, Inc.

State: NV

Country: US

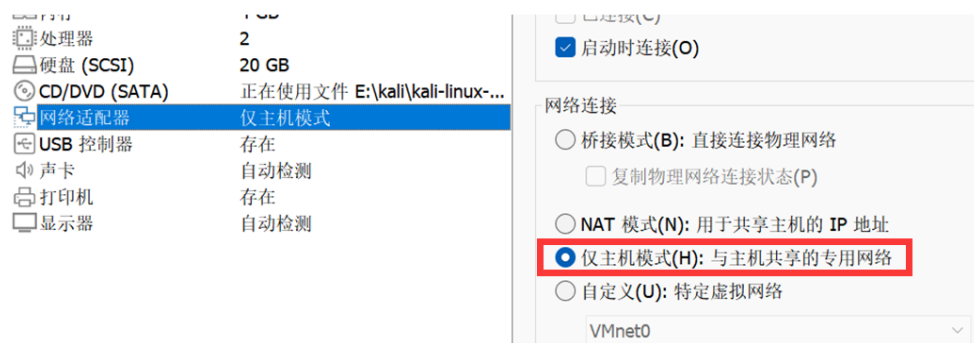
Email: [Select Request Email Form at https://domains.markmonitor.com/whois/alexa.com](https://domains.markmonitor.com/whois/alexa.com)

Raw Whois Data

Domain Name: alexa.com
Registry Domain ID: 5324166_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-08-02T02:17:33+0000
Creation Date: 1996-07-17T04:00:00+0000
Registrar Registration Expiration Date: 2025-07-16T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com

(3) 使用 Nmap 完成对目标网络中在线主机的扫描探测，并通过网络协议分析程序 Wireshark 捕获数据包，验证 ARP 主机扫描探测和 ICMP 主机扫描探测技术，理解防火墙防止扫描探测的重要作用。

- 一台宿主机和两台虚拟机组建局域网（虚拟机进行网络配置时选择 HostOnly 模式）



使用三台虚拟机互相进行攻防，使用仅主机模式是构建一个封闭环境，虚拟机只能和主机之间进行通信，此时虚拟机是无法连接上外网的，这样实验结果就仅会与虚拟机之间与虚拟机和主机之间的通信有关，而与网络中的其他主机无关，使用桥接模式理论上也许，但虚拟机可以访问外网，此时 Nmap 和 Wireshark 的运行结果很可能会受到网络中其他主机以及其他主机之间的通信的影响。

- 获取各虚拟机 ip 地址，测试相互可 ping。

```
(root@kali)-[~]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.121.129 netmask 255.255.255.0 broadcast 192.168.121.255
    ether 00:0c:29:33:c5:d6 txqueuelen 1000 (Ethernet)
    RX packets 747 bytes 86012 (83.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 762 bytes 67683 (66.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 19 base 0x2000
```

```
C:\Users\26032>ping 192.168.121.129

正在 Ping 192.168.121.129 具有 32 字节的数据:
来自 192.168.121.129 的回复: 字节=32 时间<1ms TTL=64
来自 192.168.121.129 的回复: 字节=32 时间<1ms TTL=64
来自 192.168.121.129 的回复: 字节=32 时间<1ms TTL=64
来自 192.168.121.129 的回复: 字节=32 时间<1ms TTL=64

192.168.121.129 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 0ms, 最长 = 0ms, 平均 = 0ms
```

(其他以此类推，windows 获取 ip 地址为 ipconfig)

- 攻击主机上安装 Nmap 和 Wireshark

安装 nmap: `sudo apt-get install nmap`

```
(root@kali)-[~]
# nmap
Nmap 7.93 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
```

主机 ARP 扫描（扫描网络中存在的主机）

在 Filter 中设置“arp”，只捕获 arp 包，在 Wireshark 上选择“捕获”→“开始”



使用 ping 模式进行主机发现，nmap -sP 192.168.121.1/24，进行网段下的主机扫描。

```
(root@kali)-[~]
# nmap -sP 192.168.121.1/24
Starting Nmap 7.93 ( https://nmap.org ) at 2024-09-20 14:00 CST
Nmap scan report for 192.168.121.1
Host is up (0.0012s latency).
MAC Address: 00:50:56:C0:00:01 (VMware)
Nmap scan report for 192.168.121.128 win10
Host is up (0.00020s latency).
MAC Address: 00:0C:29:9D:80:C1 (VMware)
Nmap scan report for 192.168.121.130 winxp
Host is up (0.00093s latency).
MAC Address: 00:0C:29:F5:14:36 (VMware)
Nmap scan report for 192.168.121.254 服务器端
Host is up (0.00028s latency).
MAC Address: 00:50:56:FE:73:4B (VMware)
Nmap scan report for 192.168.121.129 扫描端
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 27.94 seconds
```

其中也包括 1 为物理机的虚拟网卡 IP，129 为扫描端 IP（因为扫描端和服务器端在同一网段下），254 为服务器端。

- 在 Wireshark 上选择“Capture”→“Stop”停止嗅探，查看、记录并分析嗅探结果。

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.1? Tell 192.168.121.129
2	0.000076736	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.2? Tell 192.168.121.129
3	0.000111301	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.3? Tell 192.168.121.129
4	0.000182176	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.4? Tell 192.168.121.129
5	0.000232502	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.5? Tell 192.168.121.129
6	0.000282767	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.6? Tell 192.168.121.129
7	0.000330127	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.7? Tell 192.168.121.129
8	0.000374672	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.8? Tell 192.168.121.129
9	0.000416682	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.9? Tell 192.168.121.129
10	0.000471085	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.10? Tell 192.168.121.129
11	0.000670301	VMware_c0:00:01	VMware_33:c5:d6	ARP	68	192.168.121.1 is at 00:50:56:c0:00:01
12	0.003217544	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.13? Tell 192.168.121.129
13	0.003265666	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.14? Tell 192.168.121.129
14	0.100813511	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.17? Tell 192.168.121.129
15	0.100900977	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.18? Tell 192.168.121.129
16	0.100944840	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.19? Tell 192.168.121.129
17	0.101048638	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.20? Tell 192.168.121.129
18	0.101119983	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.21? Tell 192.168.121.129
19	0.101190447	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.22? Tell 192.168.121.129
20	0.101277843	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.23? Tell 192.168.121.129
21	0.101389626	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.24? Tell 192.168.121.129
22	0.101499044	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.25? Tell 192.168.121.129
23	0.103846736	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.28? Tell 192.168.121.129
24	0.103903394	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.29? Tell 192.168.121.129
25	0.201584885	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.32? Tell 192.168.121.129
26	0.201678182	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.33? Tell 192.168.121.129
27	0.201712938	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.34? Tell 192.168.121.129
28	0.201741052	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.35? Tell 192.168.121.129
29	0.201855359	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.36? Tell 192.168.121.129
30	0.201867402	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.37? Tell 192.168.121.129
31	0.201904432	VMware_33:c5:d6	Broadcast	ARP	42	42 Who has 192.168.121.38? Tell 192.168.121.129

通过扫描端 wireshark 抓包，可以看到扫描端发送 ARP 广播，向该网段所有可能 IP 广播询问这些 IP 对应的 MAC 地址为多少，如果收到了其他 IP 的相应就说明该 IP 存在。

● Arp 请求包

```

Frame 40: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0
Section number: 1
Interface id: 0 (eth0)
Interface name: eth0
Encapsulation type: Ethernet (1)
Arrival Time: Sep 20, 2024 14:06:14.466348762 CST
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1726812374.466348762 seconds
[Time delta from previous captured frame: 0.000080693 seconds]
[Time delta from previous displayed frame: 0.000080693 seconds]
[Time since reference or first frame: 0.303008366 seconds]
Frame Number: 40
Frame Length: 42 bytes (336 bits)
Capture Length: 42 bytes (336 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:arp]
[Coloring Rule Name: ARP]
[Coloring Rule String: arp]

Ethernet II, Src: VMware_33:c5:d6 (00:0c:29:33:c5:d6), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Destination: Broadcast (ff:ff:ff:ff:ff:ff)
Address: Broadcast (ff:ff:ff:ff:ff:ff)
.....1. .... = LG bit: Locally administered address (this is NOT the f...
.....1. .... = IG bit: Group address (multicast/broadcast)
Source: VMware_33:c5:d6 (00:0c:29:33:c5:d6)
Address: VMware_33:c5:d6 (00:0c:29:33:c5:d6)
.....0. .... = LG bit: Globally unique address (factory default)
.....0. .... = IG bit: Individual address (unicast)
Type: ARP (0x0806)
Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: VMware_33:c5:d6 (00:0c:29:33:c5:d6)
Sender IP address: 192.168.121.129
Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
Target IP address: 192.168.121.50

```

主机 ICMP 扫描

◆ 在 Filter 中设置 “icmp”，只捕获 icmp 包。开关防火墙，使用 ping 192.168.121.128 观察主机。

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开启防火墙：

No.	Time	Source	Destination	Protocol	Length	Info
2	0.000009458	192.168.121.1	192.168.121.128	ICMP	111	Destination unreachable (Port unreachable)
4	0.019224762	192.168.121.1	192.168.121.128	ICMP	111	Destination unreachable (Port unreachable)
6	0.198861691	192.168.121.1	192.168.121.128	ICMP	109	Destination unreachable (Port unreachable)
8	0.222532252	192.168.121.1	192.168.121.128	ICMP	109	Destination unreachable (Port unreachable)
9	1.014090819	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=1/256, ttl=64 (no reply)
11	1.019248564	192.168.121.1	192.168.121.128	ICMP	111	Destination unreachable (Port unreachable)
13	1.238090783	192.168.121.1	192.168.121.128	ICMP	109	Destination unreachable (Port unreachable)
14	2.034182865	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=2/512, ttl=64 (no reply)
16	2.966096103	192.168.121.1	192.168.121.128	ICMP	117	Destination unreachable (Port unreachable)
18	2.987791529	192.168.121.1	192.168.121.128	ICMP	117	Destination unreachable (Port unreachable)
20	3.019503353	192.168.121.1	192.168.121.128	ICMP	111	Destination unreachable (Port unreachable)
21	3.057597634	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=3/768, ttl=64 (no reply)
23	3.238246626	192.168.121.1	192.168.121.128	ICMP	109	Destination unreachable (Port unreachable)
25	3.988285726	192.168.121.1	192.168.121.128	ICMP	117	Destination unreachable (Port unreachable)
26	4.081646465	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=4/1024, ttl=64 (no reply)
31	5.105553206	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=5/1280, ttl=64 (no reply)
33	6.003562126	192.168.121.1	192.168.121.128	ICMP	117	Destination unreachable (Port unreachable)
34	6.129834872	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=6/1536, ttl=64 (no reply)
36	7.034908006	192.168.121.1	192.168.121.128	ICMP	111	Destination unreachable (Port unreachable)
37	7.154049560	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0001, seq=7/1792, ttl=64 (no reply)
39	7.253701969	192.168.121.1	192.168.121.128	ICMP	109	Destination unreachable (Port unreachable)

关闭防火墙：

No.	Time	Source	Destination	Protocol	Length	Info
0	0.000000000	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=1/256, ttl=64 (reply in 2)
027083836	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=2/512, ttl=64 (reply in 4)
051145277	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=3/768, ttl=64 (reply in 6)
052849144	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=4/1024, ttl=64 (reply in 8)
067483296	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=5/1280, ttl=64 (reply in 10)
091300029	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=6/1536, ttl=64 (reply in 14)
138591740	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=7/1792, ttl=64 (reply in 18)
140712070	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=8/2048, ttl=64 (reply in 20)
142719733	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=9/2304, ttl=64 (reply in 22)
155362425	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=10/2560, ttl=64 (reply in 24)
0.179040903	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=11/2816, ttl=64 (reply in 26)
1.203643762	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=12/3072, ttl=64 (reply in 28)
2.205186091	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=13/3328, ttl=64 (reply in 30)
3.206690738	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=14/3584, ttl=64 (reply in 32)
4.211419801	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=15/3840, ttl=64 (reply in 34)
5.235051972	192.168.121.129	192.168.121.128	192.168.121.128	ICMP	98	Echo (ping) request id=0x0002, seq=16/4096, ttl=64 (reply in 36)
000790736	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=1/256, ttl=128 (request in 1)
027553199	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=2/512, ttl=128 (request in 3)
052433978	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=3/768, ttl=128 (request in 5)
053437542	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=4/1024, ttl=128 (request in 7)
067961405	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=5/1280, ttl=128 (request in 9)
091788107	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=6/1536, ttl=128 (request in 13)
139824646	192.168.121.128	192.168.121.129	192.168.121.128	ICMP	98	Echo (ping) reply id=0x0002, seq=7/1792, ttl=128 (request in 17)

Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0

Section number: 1

Interface id: 0 (eth0)

Interface name: eth0

Encapsulation type: Ethernet (1)

Arrival Time: Sep 20, 2024 14:18:06.919488781 (CST)

[Time shift for this packet: 0.000000000 seconds]

Epoch Time: 1726813086.919488781 seconds

[Time delta from previous captured frame: 0.000000000 seconds]

[Time delta from previous displayed frame: 0.000000000 seconds]

[Time since reference or first frame: 0.000000000 seconds]

Frame Number: 1

Frame Length: 98 bytes (784 bits)

Capture Length: 98 bytes (784 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:icmp:data]

[Coloring Rule Name: ICMP]

[Coloring Rule String: icmp || icmpv6]

Ethernet II, Src: VMware_33:c5:d6 (00:0c:29:9d:80:c1), Destination: VMware_9d:80:c1 (00:0c:29:9d:80:c1)

Address: VMware_9d:80:c1 (00:0c:29:9d:80:c1)

... ..0. = LG bit: Glob...

... ..0. = IG bit: Indi...

Source: VMware_33:c5:d6 (00:0c:29:33:c5:d6)

root@kali:~# ping 192.168.121.128

PING 192.168.121.128 (192.168.121.128) 56(84) bytes of data:

64 bytes from 192.168.121.128: icmp_seq=1 ttl=128 time=0.848 ms

64 bytes from 192.168.121.128: icmp_seq=2 ttl=128 time=0.490 ms

64 bytes from 192.168.121.128: icmp_seq=3 ttl=128 time=1.32 ms

64 bytes from 192.168.121.128: icmp_seq=4 ttl=128 time=0.613 ms

64 bytes from 192.168.121.128: icmp_seq=5 ttl=128 time=0.502 ms

64 bytes from 192.168.121.128: icmp_seq=6 ttl=128 time=0.513 ms

64 bytes from 192.168.121.128: icmp_seq=7 ttl=128 time=1.27 ms

64 bytes from 192.168.121.128: icmp_seq=8 ttl=128 time=0.946 ms

64 bytes from 192.168.121.128: icmp_seq=9 ttl=128 time=0.674 ms

64 bytes from 192.168.121.128: icmp_seq=10 ttl=128 time=0.418 ms

64 bytes from 192.168.121.128: icmp_seq=11 ttl=128 time=0.627 ms

64 bytes from 192.168.121.128: icmp_seq=12 ttl=128 time=0.675 ms

64 bytes from 192.168.121.128: icmp_seq=13 ttl=128 time=0.721 ms

64 bytes from 192.168.121.128: icmp_seq=14 ttl=128 time=0.577 ms

64 bytes from 192.168.121.128: icmp_seq=15 ttl=128 time=0.568 ms

64 bytes from 192.168.121.128: icmp_seq=16 ttl=128 time=0.442 ms

— 192.168.121.128 ping statistics —

16 packets transmitted, 16 received, 0% packet loss, time 15235ms

rtt min/avg/max/mdev = 0.418/0.700/1.318/0.262 ms

(4) 使用 Nmap 完成对主机操作系统类型的探测和端口的扫描探测, 并通过 Wireshark 捕获扫描数据包, 验证 Nmap 所使用的 SYN 扫描探测技术。

- 在 Filter 中设置 `ip.addr==192.168.121.129`, 只捕获本机与目标机通信的数据包, 消除无关数据对结果的影响 (查看三次握手情况) 通过 Nmap 对主机进行端口扫描
- 在 cmd 命令行使用命令: `nmap -sS 192.168.121.128`, 进行主机端口扫描, 查看并记录扫描结果。

No.	Time	Source	Destination	Protocol	Length	Info
2246	14.392099007	192.168.121.129	192.168.121.128	TCP	60	222 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2245	14.390472664	192.168.121.129	192.168.121.128	TCP	58	53154 → 222 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2244	14.390220915	192.168.121.129	192.168.121.128	TCP	60	27352 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2243	14.390216928	192.168.121.129	192.168.121.128	TCP	60	3800 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2242	14.390200507	192.168.121.129	192.168.121.128	TCP	60	541 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2241	14.388025461	192.168.121.129	192.168.121.128	TCP	58	53154 → 27352 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2240	14.388010242	192.168.121.129	192.168.121.128	TCP	58	53154 → 3800 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2239	14.387996297	192.168.121.129	192.168.121.128	TCP	58	53154 → 541 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2238	14.387918828	192.168.121.129	192.168.121.128	TCP	60	2725 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2237	14.387906435	192.168.121.129	192.168.121.128	TCP	60	9418 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2236	14.386160465	192.168.121.129	192.168.121.128	TCP	58	53154 → 2725 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2235	14.386133153	192.168.121.129	192.168.121.128	TCP	58	53154 → 9418 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2234	14.376005066	192.168.121.129	192.168.121.128	TCP	60	4279 → 53154 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2233	14.373243610	192.168.121.129	192.168.121.128	TCP	58	53154 → 4279 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2232	14.370775041	192.168.121.129	192.168.121.128	TCP	60	38292 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2231	14.370772376	192.168.121.129	192.168.121.128	TCP	60	4550 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2230	14.370769139	192.168.121.129	192.168.121.128	TCP	60	2045 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2229	14.370765803	192.168.121.129	192.168.121.128	TCP	60	5963 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2228	14.370762226	192.168.121.129	192.168.121.128	TCP	60	50002 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2227	14.370754882	192.168.121.129	192.168.121.128	TCP	60	714 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2226	14.370563629	192.168.121.129	192.168.121.128	TCP	60	7999 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2225	14.370560483	192.168.121.129	192.168.121.128	TCP	60	32772 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2224	14.370557177	192.168.121.129	192.168.121.128	TCP	60	1031 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2223	14.370554131	192.168.121.129	192.168.121.128	TCP	60	2111 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2222	14.370550414	192.168.121.129	192.168.121.128	TCP	60	2144 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2221	14.370534413	192.168.121.129	192.168.121.128	TCP	60	6779 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2220	14.369826307	192.168.121.129	192.168.121.128	TCP	58	53152 → 38292 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2219	14.369812551	192.168.121.129	192.168.121.128	TCP	58	53152 → 4550 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2218	14.369798515	192.168.121.129	192.168.121.128	TCP	58	53152 → 2045 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2217	14.369770792	192.168.121.129	192.168.121.128	TCP	58	53152 → 5963 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2216	14.369756946	192.168.121.129	192.168.121.128	TCP	58	53152 → 50002 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2215	14.369742729	192.168.121.129	192.168.121.128	TCP	58	53152 → 714 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2214	14.369728822	192.168.121.129	192.168.121.128	TCP	58	53152 → 7999 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2213	14.369714405	192.168.121.129	192.168.121.128	TCP	58	53152 → 32772 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2212	14.369700629	192.168.121.129	192.168.121.128	TCP	58	53152 → 1031 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2211	14.369686542	192.168.121.129	192.168.121.128	TCP	58	53152 → 2111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2210	14.369679561	192.168.121.129	192.168.121.128	TCP	58	53152 → 2144 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2209	14.369645263	192.168.121.129	192.168.121.128	TCP	58	53152 → 6779 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2208	14.367010540	192.168.121.129	192.168.121.128	TCP	60	2005 → 53152 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

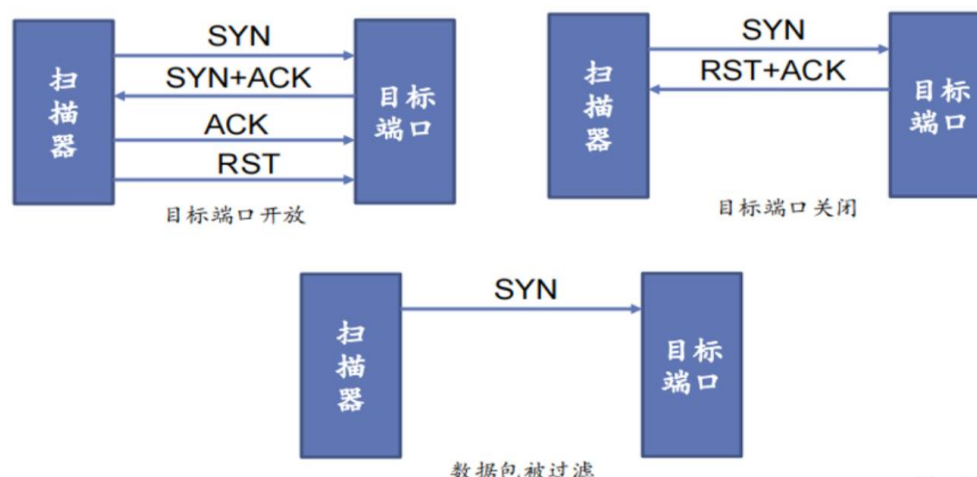
如果只是单个的一个 SYN, 它表示的只是建立连接。SYN 与 FIN 是不会同时为 1 的, 因为前者表示的是建立连接, 而后者表示的是断开连接。RST 一般是在 FIN 之后才会出现为 1 的情况, 表示的是连接重置。

一般地, 当出现 FIN 包或 RST 包时, 我们便认为客户端与服务器端断开了连接; 而当出现 SYN 和 SYN+ACK 包时, 我们认为客户端与服务器建立了一个连接。

分析:

`nmap -sS 192.168.121.128` 进行的是 SYN 扫描, SYN 扫描相对来说不张扬, 不易被注意到, 因为它从来不完全完成 TCP 连接, 它常常被称为半开放扫描, 因为它不打开一个完全的 TCP 连接。它发送一个 SYN 报文, 就像您真的要打开一个连接, 然后等待响应。SYN/ACK 表示端口在监听 (开放), 而 RST (复位) 表示没有监听者。

通过抓包可以看出, Nmap 会向所有端口发送 SYN 包, 由此查看该端口是否开放, 如果目标端口是开放的, 那么目标系统通常会响应一个 SYN-ACK 包。此时, Nmap 不会完成 TCP 三次握手的第三个步骤, 而是发送一个 RST (重置) 包来终止这个半开放连接。如果目标端口关闭, 通常会收到一个 RST 包作为响应。如果 Nmap 没有收到任何响应, 或者收到了 ICMP 不可达消息, 那么它可能会将该端口标记为被过滤 (filtered)。



解决方法:

进行 TCP connect()扫描: nmap -sT 192.168.121.128。

● 发现三次握手:

287	17.939612391	192.168.121.129	192.168.121.128	TCP	74 57800 → 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_Pe
288	17.939688105	192.168.121.129	192.168.121.128	TCP	74 42732 → 1024 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_Pe
289	17.939708436	192.168.121.129	192.168.121.128	TCP	66 445 → 38350 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=7
290	17.939717863	192.168.121.129	192.168.121.128	TCP	54 38350 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0

第一次握手: 客户端发送一个 TCP, 标志位为 SYN, 序列号为 0, 代表客户端请求建立连接。

```

Transmission Control Protocol, Src Port: 42732, Dst Port: 1024, Seq: 0, Len: 0
  Source Port: 42732
  Destination Port: 1024
  [Stream index: 28]
  [Conversation completeness: Incomplete (37)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number) 序列号为0
  Sequence Number (raw): 2030633298
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1010 .... = Header Length: 40 bytes (10)
  Flags: 0x002 (SYN) 标志位为SYN
  Window: 64240
  [Calculated window size: 64240]
  Checksum: 0x18f9 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), 1
  [Timestamps]
    
```

第二次握手: 服务器发回确认包,标志位为 SYN,ACK.将确认序号(AcknowledgementNumber)设置为客户的 ISN 加 1 以.即 0+1=1。

```

Transmission Control Protocol, Src Port: 445, Dst Port: 38350, Seq: 0, Ack: 1, Len: 0
  Source Port: 445
  Destination Port: 38350
  [Stream index: 25]
  [Conversation completeness: Complete, NO_DATA (39)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number) 序列号为0
  Sequence Number (raw): 1835519394
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number) 收到确认包
  Acknowledgment number (raw): 2087950671
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x012 (SYN, ACK) 标志位为SYN、ACK TCP第二次握手
  Window: 65535
  [Calculated window size: 65535]
  Checksum: 0x1255 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (1
  [Timestamps]
  [SEQ/ACK analysis]
    
```

第三次握手：客户端再次发送确认包(ACK)SYN 标志位为 0,ACK 标志位为 1.并且把服务器发来的 ACK 的序号字段+1,放在确定字段中发送给对方.并且在数据段放写 ISN 的+1

```

Destination Address: 192.168.121.128
Transmission Control Protocol, Src Port: 38350, Dst Port: 445, Seq: 1, Ack: 1, Len: 0
Source Port: 38350
Destination Port: 445
[Stream index: 25]
[Conversation completeness: Complete, NO_DATA (39)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2087950671
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1835519395
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 502
[Calculated window size: 64256]
[Window size scaling factor: 128]
Checksum: 0x5132 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
    
```

进行端口扫描

```

(root@kali)-[~]
# nmap -sS 192.168.121.128
Starting Nmap 7.93 ( https://nmap.org ) at 2024-09-20 14:27 CST
Nmap scan report for 192.168.121.128
Host is up (0.0018s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 00:0C:29:9D:80:C1 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 14.50 seconds
    
```

Win10 机上 135、139、445 端口是开放的。

No.	Time	Source	Destination	Protocol	Length	Info
213	15.960736891	192.168.121.129	192.168.121.128	TCP	58	45357 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
214	15.960991114	192.168.121.129	192.168.121.128	TCP	58	45357 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
215	15.961161478	192.168.121.128	192.168.121.129	TCP	60	1723 → 45357 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
216	15.961829310	192.168.121.128	192.168.121.129	TCP	60	445 → 45357 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
217	15.961842986	192.168.121.129	192.168.121.128	TCP	54	45357 → 445 [RST] Seq=1 Win=0 Len=0
218	15.962063235	192.168.121.129	192.168.121.128	TCP	58	45357 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
219	15.962273224	192.168.121.129	192.168.121.128	TCP	58	45357 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
220	15.962311847	192.168.121.129	192.168.121.128	TCP	58	45357 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
221	15.962343197	192.168.121.129	192.168.121.128	TCP	58	45357 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
222	15.962375588	192.168.121.129	192.168.121.128	TCP	58	45357 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
223	15.962413240	192.168.121.129	192.168.121.128	TCP	58	45357 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
224	15.962455290	192.168.121.129	192.168.121.128	TCP	58	45357 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
225	15.962620203	192.168.121.128	192.168.121.129	TCP	60	110 → 45357 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
226	15.962627597	192.168.121.128	192.168.121.129	TCP	60	3389 → 45357 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
227	15.962630202	192.168.121.128	192.168.121.129	TCP	60	80 → 45357 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
228	15.962632777	192.168.121.128	192.168.121.129	TCP	60	1025 → 45357 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

可以看到 kali 向被扫描主机的各个端口发起 TCP 请求以探测端口开放情况。

操作系统类型探测

- ◆ nmap -O 192.168.121.128, 进行主机端口扫描, 查看并记录扫描结果。


```
(root@kali)~[~]
# sudo nmap -O 192.168.121.128
Starting Nmap 7.93 ( https://nmap.org ) at 2024-09-20 15:30 CST
Nmap scan report for 192.168.121.128
Host is up (0.00048s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:9D:80:C1 (VMware)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1709 - 1909
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 15.67 seconds
```

四、讨论与分析

思考题：Nmap 和 Wireshark 在网络管理工作中可以给予我们哪些帮助？

在网络安全网络资源信息收集实验中，Nmap 和 Wireshark 是两个非常重要的工具，它们在网络管理工作中可以提供以下帮助：

Nmap

1. 网络扫描：

- Nmap 可以快速扫描网络，识别活动的主机、开放的端口和运行的服务。这对于了解网络拓扑和设备状态非常重要。

2. 服务识别：

- Nmap 能够识别主机上运行的服务及其版本信息，帮助网络管理员了解潜在的安全漏洞。

3. 操作系统检测：

- Nmap 可以通过分析响应数据包来推测目标主机的操作系统类型和版本，这对于制定相应的安全策略非常有用。

4. 安全审计：

- 通过定期使用 Nmap 进行扫描，网络管理员可以发现未授权的设备或服务，及时进行安全审计和风险评估。

Wireshark

1. 数据包捕获：

- Wireshark 是一个强大的网络协议分析工具，可以实时捕获和分析网络流量，帮助管理员监控网络活动。

2. 流量分析：

- 通过分析捕获的数据包，Wireshark 可以帮助识别网络中的异常流量、潜在的攻击行为

或性能瓶颈。

3. 故障排除：

- Wireshark 可以帮助网络管理员诊断网络问题，通过查看数据包的详细信息，找出故障的根源。

4. 协议分析：

- Wireshark 支持多种网络协议的分析，管理员可以深入了解网络通信的细节，优化网络配置和性能。

结合 Nmap 和 Wireshark，网络管理员可以全面了解网络环境，及时发现和解决安全隐患，提升网络的安全性和稳定性。这两个工具在网络管理和安全防护中扮演着不可或缺的角色。

五、实验者自评（从实验设计、实验过程、对实验知识点的理解上给出客观公正的自我评价）

1. 实验设计

目标明确性：实验的目标设定清晰，旨在通过使用 Nmap 和 Wireshark 进行网络资源的有效收集与分析，帮助理解网络安全的基本概念和工具的应用。

方法合理性：选择了合适的工具（Nmap 和 Wireshark）进行实验，设计了合理的实验步骤，包括网络扫描、数据包捕获和分析，确保了实验的系统性和科学性。

预期结果：预期能够识别网络中的设备、服务及其安全状态，分析网络流量并识别潜在的安全威胁，设计的实验能够有效达成这些目标。

2. 实验过程

执行情况：在实验过程中，能够按照设计步骤有序进行，成功使用 Nmap 进行网络扫描，获取了目标网络的基本信息，并使用 Wireshark 捕获了网络流量。

问题处理：在实验中遇到了一些技术问题，例如网络配置不当导致扫描失败，但能够及时调整设置并重新进行实验，体现了较强的问题解决能力。

数据记录：实验过程中对数据的记录和整理较为规范，能够清晰地展示扫描结果和流量分析，便于后续的分析 and 总结。

3. 对实验知识点的理解

理论知识：对 Nmap 和 Wireshark 的基本功能和使用方法有了深入的理解，能够解释各个功能模块的作用及其在网络安全中的重要性。

实践应用：通过实际操作，理解了网络扫描和流量分析的实际应用场景，能够将理论知识与实践相结合，提升了对网络安全的整体认识。

反思与改进：在实验结束后，能够反思

实验过程中的不足之处，例如对某些高级功能的掌握不够深入，未来需要进一步学习和实践，以提升自己的网络安全技能。

总结

总体而言，本次实验设计合理，过程顺利，知识点理解到位。通过此次实验，不仅提升了对网络安全工具的使用能力，也增强了对网络安全整体概念的理解。未来将继续深入学习相关知识，提升自己的实践能力和理论水平。

六、附录：关键代码（给出适当注释，可读性高）

无